

INTERNSHIP PROJECT REPORT

California House Price Prediction Using Linear Regression

A Supervised Machine Learning Approach to Predicting Median House Values

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Internship: Artificial Intelligence & Machine Learning	Dataset: California Housing Dataset (scikit-learn)

1. Introduction & Objective

This report documents Task 1 of an internship in Artificial Intelligence and Machine Learning: the design, implementation, and evaluation of a Linear Regression model to predict median house values across California. Linear Regression is one of the most fundamental supervised learning algorithms and provides an interpretable baseline for understanding the relationship between a set of input features and a continuous target variable, making it a natural starting point for a regression-based machine learning task.

The objective of this task is to build and evaluate a Linear Regression model that predicts the median house value of a California district using eight socio-economic and geographic features, and to assess the model's predictive performance using standard regression evaluation metrics.

2. Dataset Description

The California Housing dataset is a widely used benchmark dataset for regression tasks, obtained directly through scikit-learn's `fetch_california_housing` utility. It is derived from the 1990 U.S. Census and contains housing and demographic information aggregated at the census block-group level across California.

Property	Value
Total Records	20,640 block groups
Total Features	8 independent features + 1 target variable
Missing Values	None (fully complete dataset)
Target Variable	MedHouseVal — Median House Value (in \$100,000s)

Table 1: Dataset overview

2.1 Feature Description

Feature	Description
MedInc	Median income of households within a block group (in tens of thousands of USD)
HouseAge	Median age of houses within a block group (years)
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Total population within a block group
AveOccup	Average number of household members (occupancy rate)
Latitude	Geographic latitude of the block group
Longitude	Geographic longitude of the block group
MedHouseVal	Target: median house value for households (in \$100,000s)

Table 2: Feature descriptions

2.2 Descriptive Statistics

Feature	Mean	Std Dev	Min	Max
MedInc	3.87	1.90	0.50	15.00

HouseAge	28.64	12.59	1.00	52.00
AveRooms	5.43	2.47	0.85	141.91
AveBedrms	1.10	0.47	0.33	34.07
Population	1,425.48	1,132.46	3.00	35,682.00
AveOccup	3.07	10.39	0.69	1,243.33
Latitude	35.63	2.14	32.54	41.95
Longitude	-119.57	2.00	-124.35	-114.31
MedHouseVal	2.07	1.15	0.15	5.00

Table 3: Descriptive statistics of dataset features (n = 20,640)

3. Methodology / ML Workflow

The project follows a structured end-to-end machine learning workflow, implemented in Python using NumPy, Pandas, Matplotlib, Seaborn, and scikit-learn:

1. Data Loading — importing the California Housing dataset via scikit-learn
2. Data Exploration — inspecting shape, data types, and missing values
3. Exploratory Data Analysis (EDA) — visualizing distributions and feature correlations
4. Data Preprocessing — separating features and target, checking data quality
5. Train-Test Split — dividing data into training (80%) and testing (20%) subsets
6. Feature Scaling — standardizing features using StandardScaler
7. Model Training — fitting a Linear Regression model on the training data
8. Prediction — generating predictions on unseen test data
9. Model Evaluation — assessing performance using MAE, RMSE, and R^2
10. Visualization & Conclusion — analyzing results through plots and summarizing findings

4. Exploratory Data Analysis

Exploratory Data Analysis was carried out to understand the distribution of the target variable and the relationships between features prior to model building.

4.1 Distribution of Median House Values

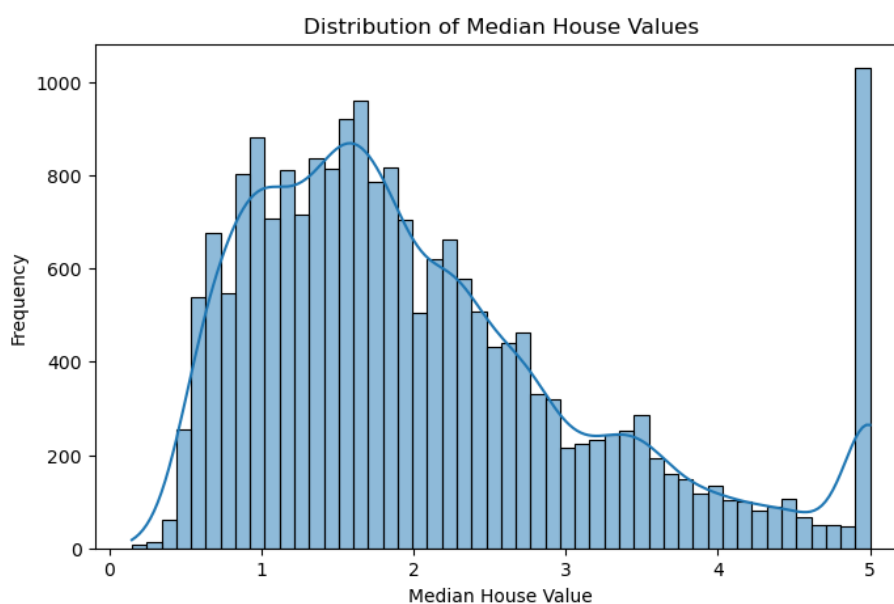


Figure 1: Distribution of Median House Values

The target variable is right-skewed, with most district house values concentrated below \$300,000. A distinct spike is visible at the upper boundary near \$500,000, corresponding to the dataset's known capping of house values at this ceiling.

4.2 Correlation Analysis

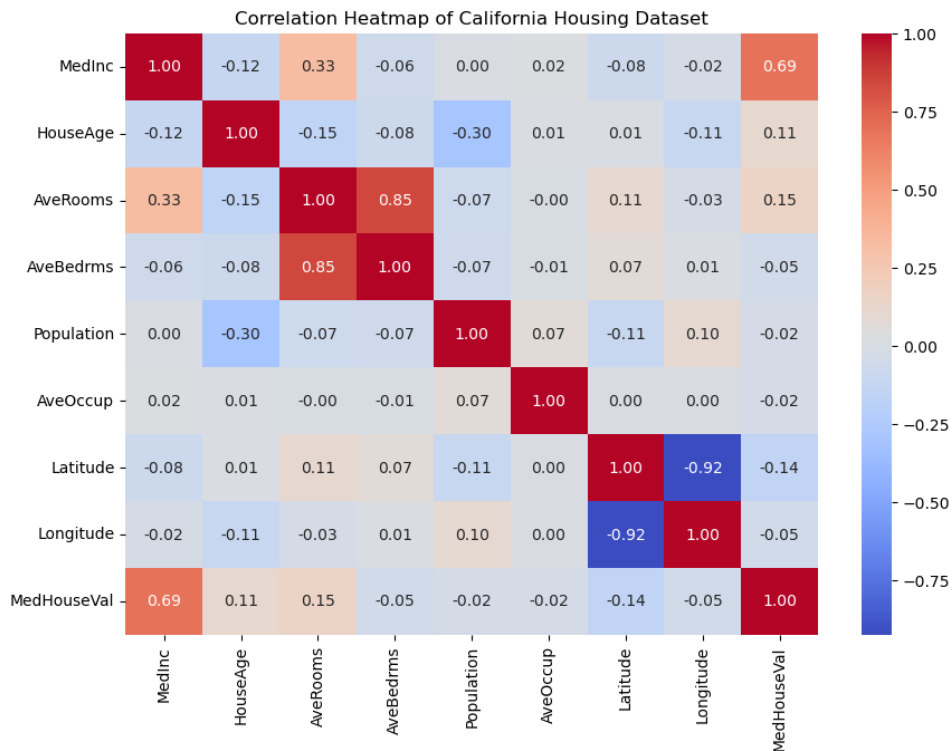


Figure 2: Correlation Heatmap of the California Housing Dataset

The correlation heatmap shows that MedInc (median income) has the strongest positive correlation with MedHouseVal, at approximately 0.69, indicating that median income is the most influential predictor of house value in this dataset. AveRooms and AveBedrms are strongly correlated with each other (0.85), suggesting potential multicollinearity between these two features. Latitude and Longitude show a strong negative correlation (-0.92), which reflects the geographic orientation of California rather than a direct predictive relationship.

4.3 Median Income vs Median House Value



Figure 3: Median Income vs Median House Value

The scatter plot confirms a clear positive relationship between median income and median house value: districts with higher median income tend to have higher median house values. The horizontal band of points at the \$500,000 ceiling further illustrates the capping effect observed in the target variable's distribution.

5. Data Preprocessing

Before model training, the dataset was prepared through feature-target separation, a train-test split, and feature scaling.

5.1 Feature–Target Separation

The dataset was split into a feature matrix X (all columns except MedHouseVal, shape $20,640 \times 8$) and a target vector y (MedHouseVal, shape 20,640).

5.2 Train-Test Split

Subset	Samples	Proportion
Training Set	16,512	80%
Testing Set	4,128	20%

Table 4: Train-test split (random_state = 42)

5.3 Feature Scaling

Since the raw features span very different numerical ranges (e.g., Population reaches into the tens of thousands while AveBedrms remains close to 1), StandardScaler was applied to standardize all features to zero mean and unit variance. The scaler was fitted only on the training data and then used to transform both the training and test sets, preventing data leakage from the test set.

6. Model Building

A Linear Regression model was used to model the relationship between the input features and the target variable. Linear Regression estimates the target as a weighted linear combination of the input features, of the form $\hat{y} = b_0 + b_1x_1 + b_2x_2 + \dots + b_8x_8$, where the coefficients ($b_1 \dots b_8$) are learned by minimizing the sum of squared residuals between predicted and actual values on the training data.

The model was instantiated using scikit-learn's LinearRegression() class and trained on the scaled training features ($X_{\text{train_scaled}}$) and corresponding target values (y_{train}). Once trained, the model was used to generate predictions on the held-out, scaled test set ($X_{\text{test_scaled}}$).

7. Model Evaluation

The trained model's performance was evaluated on the test set using three standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Metric	Value	Interpretation
MAE	0.5332	Avg. absolute error (~\$53,320)
RMSE	0.7456	Penalizes larger errors more
R^2 Score	0.5758	~57.6% variance explained

Table 5: Model evaluation metrics on the test set

7.1 Actual vs Predicted House Values

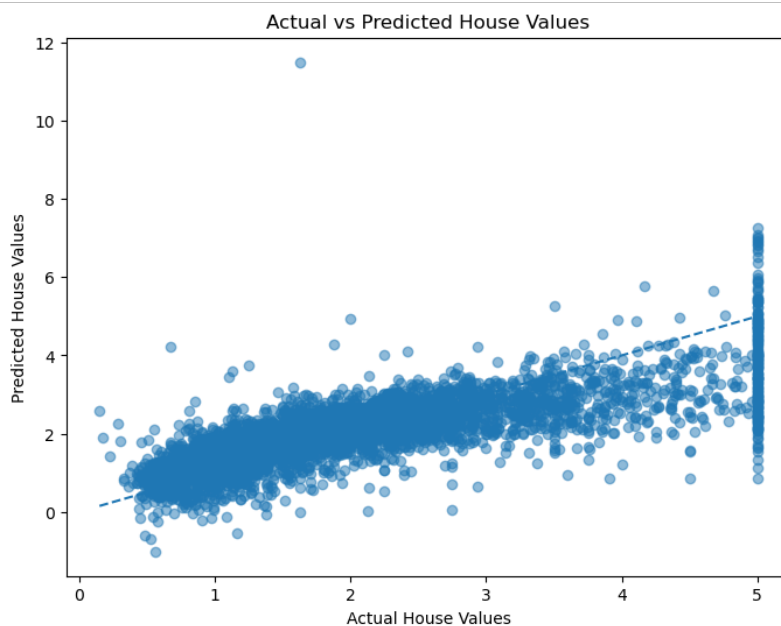


Figure 4: Actual vs Predicted House Values (dashed line = ideal fit)

Predicted values generally follow the ideal diagonal line, confirming that the model captures the overall trend in the data. However, noticeable scatter around the line — especially at higher actual values and near the \$500,000 cap — indicates that a purely linear model cannot fully capture the underlying relationship.

7.2 Residual Analysis



Figure 5: Residual Plot (Predicted Values vs Residuals)

The residual plot shows that errors are not entirely random and do not scatter uniformly around zero; a mild funnel-shaped pattern is visible, suggesting some heteroscedasticity and that a portion of the variance in house values remains unexplained by the linear model. This is consistent with the R^2 score of approximately 0.576.

8. Results & Discussion

The Linear Regression model explains roughly 57.6% of the variance in median house values, with an average absolute prediction error of about \$53,320. Median income emerged as the single strongest predictor, consistent with real-world housing market dynamics, and the model's simplicity and interpretability make it a strong, fast-training baseline.

At the same time, the visible curvature and spread in the actual-vs-predicted and residual plots indicate that non-linear relationships and feature interactions exist in the data that Linear Regression cannot capture. The capping of house values

at \$500,000 introduces artificial bias near the upper end of the target range, and multicollinearity between AveRooms and AveBedrms (correlation 0.85) may slightly affect the stability of the estimated coefficients.

9. Conclusion

In this project, a Linear Regression model was developed to predict median house values using the California Housing dataset. The complete workflow included data loading, exploration, statistical analysis, visualization, correlation analysis, feature scaling, model training, prediction, and evaluation. The model achieved an R^2 score of approximately 0.576, with an MAE of 0.5332 and an RMSE of 0.7456 — demonstrating that Linear Regression captures a significant, interpretable relationship between the selected features and house prices, while a meaningful portion of the variance remains unexplained, motivating the use of more advanced modelling techniques.

10. Future Scope

Building on this baseline, the following directions could further improve predictive performance:

- Apply regularized regression techniques (Ridge, Lasso) to address multicollinearity between correlated features
- Experiment with non-linear models such as Random Forest, Gradient Boosting, or XGBoost to capture complex feature interactions
- Engineer new features, such as rooms-per-household or bedrooms-per-room ratios, to improve signal quality
- Use k-fold cross-validation for a more robust and generalizable estimate of model performance
- Investigate and account for the capping effect on house values near the \$500,000 ceiling